

French→English Transformer from scratch: where it generalizes and where it breaks

Encoder-decoder trained from scratch on filtered opus-100; evaluated in-domain, on long sentences, and on literature it never saw.

GitHub bad2212/transformer-model

Hugging Face Badalt/fr-en-transformer-scratch

W&B run 5ks4crpz

W&B report dashboards

39.94

Dev OVERALL score

29.8 / 46.4

Seen · BLEU / chrF

34.7 / 60.0

Long · BLEU / chrF

21.9 / 43.4

Books (unseen) · BLEU / chrF

39.7M

Params · 3.6 h, one T4

KEY FINDINGS

1. Length is not the failure mode; domain is.

Literature trails in-domain at every sentence length.

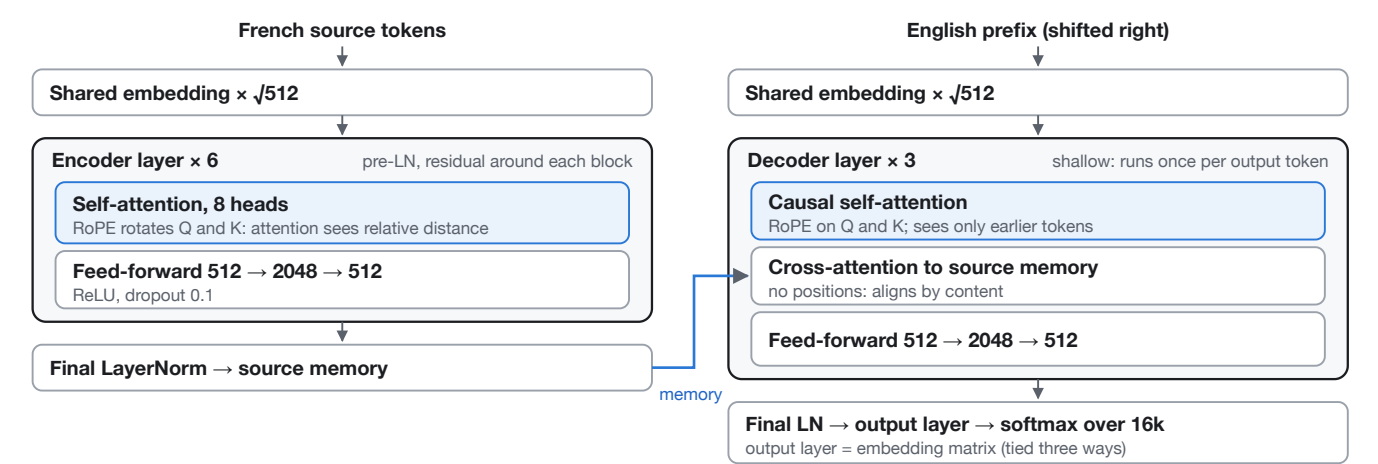
2. The dev gap needs more data to be trusted.

On 60 sentences its 95% CI includes zero; on 1,000 per domain it is 31.6 vs 15.8 BLEU.

3. Register explains more than vocabulary.

Books sentences with no rare word still trail by 14.1 BLEU; rare words cost more on top.

1 Architecture decisions



39.7M parameters. Positions enter only through RoPE inside the two self-attentions; the encoder's output (memory) feeds the cross-attention of every decoder layer; one $16k \times 512$ matrix serves as both embeddings and the output layer.

Choice	Why	Rejected alternative
Encoder-decoder [1]	Reads the source bidirectionally; cross-attention aligns target to source words	Decoder-only: causal view of the source, sequences 2× longer
6 enc / 3 dec, d 512 [2]	Measured on the T4: 6/6 ran at 11.5k target tok/s, 6/3 at 13.6k, so ~20% more updates in budget	6/6 base: slightly better per step, 15% slower
Pre-LayerNorm [3]	Stable gradients at a high peak LR (7e-4)	Post-LN: diverges easily early
RoPE, self-attention only [4]	Relative positions; cross-attention aligns by content. Not chosen for extrapolation: <i>long</i> is within the trained range	Sinusoidal (ablation config), ALiBi [5]
Tied embeddings [6]	One matrix for encoder input, decoder input and output layer; saves 16.4M params	Separate vocabularies
SentencePiece 16k [7, 8]	Joint unigram vocab, byte fallback (no <unk>), source-side subword sampling $\alpha=0.1$	BPE [9], 32k vocab, characters
Beam 5, $\alpha=1.8$ [10]	Tuned on dev; chrF ($\beta=2$) and BLEU's brevity penalty both reward full-length output	Greedy: -1.4 OVERALL
Own PyTorch code	RoPE inside attention, pre-LN, tying; ~170 lines, unit-tested	<code>nn.Transformer</code> : no RoPE hook

2 Data and training

Data. opus-100 en-fr *train* only, no subsampling. Unicode and quote normalisation (books sources use ', their references use '), then filtering kept **923,274 of 1,000,000** pairs: 32,548 duplicates, 21,617 bad length ratios, 14,770 untranslated copies, 6,602 empty, 1,186 over-long, 3 dev/test overlaps. opus_books was never used for training, the tokenizer or any tuning.

Training. Label smoothing 0.1, AdamW (0.9, 0.98), 4k warmup + inverse-sqrt, ~24k tokens per update, fp16, seed 42. Stopped at step 9,000 (~8.3 epochs, 3.6 of a planned 5.5 h) when the free-Colab GPU quota ran out, so the model is **under-trained**. Final weights average checkpoints 7k–9k [11] (greedy 38.49 vs 38.14 for 9k alone).

Step	Val loss	Dev OVERALL (greedy)		
2,000	4.11	14.4		
4,000	2.65	30.8		
6,000	2.28	35.4		
8,000	2.14	37.6		

Decoding (dev)	OVERALL	BLEU	chrF
Greedy	38.49	28.45	46.81
Beam 4, α 1.4	39.47	30.00	47.50
Beam 5, α 1.4	39.83	30.23	47.77
Beam 5, α 1.8 — chosen	39.94	30.23	47.91
Beam 5, α 2.2	39.83	30.09	47.85

Full 11-point grid in the appendix. Gains above α 1.4 are within noise.

3 Dev results by slice

Slice	n	BLEU	95% CI	chrF	95% CI
seen	60	29.8	21.9 to 38.0	46.4	40.4 to 52.4
long	30	34.7	24.9 to 43.9	60.0	53.9 to 66.1
unseen domain	60	21.9	17.1 to 26.2	43.4	39.3 to 48.0
All	150	30.2		47.9	

OVERALL = $0.4 \cdot \text{BLEU} + 0.4 \cdot \text{chrF} + 0.2 \cdot \text{chrF}_{\text{unseen}}$ = **39.94**. Bootstrap CIs use 1,000 resamples. The seen–unseen gap is 7.9 BLEU (CI –1.8 to 17.5) and 3.0 chrF (CI –4.5 to 10.6): **not significant on 60 sentences**, so section 4 re-measures it on 1,000. *Long* beats *seen* because long opus-100 sentences are formulaic official text, while short ones are colloquial subtitles with loose references (about 4 of 150 dev pairs are misaligned).

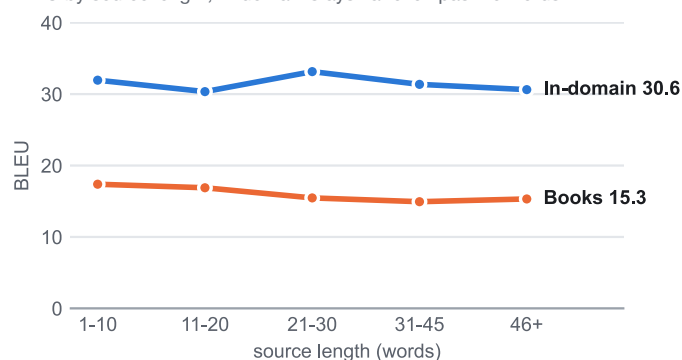
4 Generalization: evidence and hypothesis

Sample (n=1,000 each, greedy)	BLEU	chrF	Unseen words	Rare words	Ref. loss	Out/ref length
■ opus-100 test (in-domain)	31.6	52.0	1.1%	2.2%	2.05	0.94
■ opus_books sample (unseen)	15.8	37.9	1.9%	5.1%	2.88	0.90

■ In-domain (opus-100 test) ■ Books (opus_books)

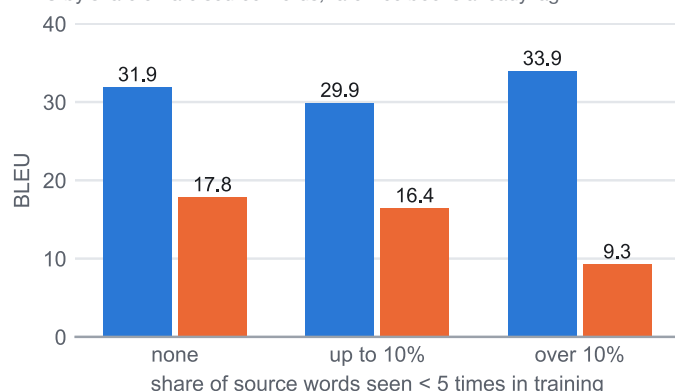
Books trail at every length

BLEU by source length; in-domain stays flat even past 45 words



Rare words hurt only in books

BLEU by share of rare source words; rare-free books already lag



- Register is the largest share.** Books sentences with no rare word score 17.8 vs 31.9 BLEU in-domain, and the model's loss on the correct reference is 2.88 vs 2.05 nats/token (perplexity 17.9 vs 7.8): trained on subtitles and official text, it finds 19th-century narrative English improbable.
- Rare vocabulary adds to it.** Books have 1.8× the never-seen-word rate and 2.4× the rare-word rate; the 190 books sentences with over 10% rare words fall to 9.3 BLEU.
- Freer references and shorter outputs.** Output/reference length is 0.90 on books vs 0.94, so BLEU's brevity penalty bites harder. Typography is ruled out by normalisation.

5 Where it fails

Failure	Examples and frequency
Names, rare words	Subword pieces recombine into a frequent word: <i>Le grand Meaulnes</i> → “The Great Meals”, <i>roulotte</i> → “roller”, <i>grognaît</i> → “grew up”
Literary word senses	The common sense wins: <i>cils</i> (eyelashes) → “cloaks”, <i>fers</i> (forceps) → “irons”. Past tenses mostly right (<i>répondit</i> → “answered”), with slips (<i>jeta</i> → “is throwing”)
Repetition loops	Rare: 0.5% of in-domain and 0.4% of books outputs repeat a 3-gram three or more times
Very long inputs	Inputs over 128 subword tokens are split at sentence ends: on 23 such sentences, chrF 41.2 split vs 39.5 unsplit
Reference noise	Misaligned opus-100 pairs and books references with stray quotes or chapter headings cap the scores

6 Challenges and how they were resolved

Problem	Resolution
Colab disconnects; GPU quota ended training at 3.6 h	Atomic full-state checkpoints, deterministic batch order, wall-clock budget across sessions, same W&B run. A planned kill-and-resume test caught a bug (RNG state moved to GPU) before the real run
T4 slower than planned	Switched 6/6 → 6/3 on a throughput threshold fixed before measuring
Noisy data; curly vs straight quotes	Filtering (7.7% removed); one normaliser on inputs and outputs (one ' costs ~20 chrF on a sentence)
Decoding stalls on one long input	Length-sorted batches capped at 2,048 source tokens; outputs unchanged
Tokenizer differs across machines	12 of 16k pieces changed with the thread count; pinned to 2 threads, and the trained tokenizer ships with the model
Beam-search and splitter edge cases	Unit tests: greedy = beam(1), overfit-and-recover, abbreviation-aware splitting

7 How the results were validated

Check	Evidence
Model correctness	5 unit tests: causal mask, padding invariance (RoPE and sinusoidal), a tiny model memorises 8 pairs and greedy, beam(1) and beam(4) all recover them, normalisation, sentence splitting
Metric fidelity	Scoring imports the official <code>score.py</code> ; identical OVERALL (39.94) to running it directly
No leakage	Dev/test sources removed from training; opus_books used only after every choice was fixed; all tuning on dev
Reproducibility	Identical filter counts on two machines; seeded batch order; one command reproduces the run
Uncertainty	Bootstrap CIs on every dev slice; the gap re-measured on 1,000 sentences per domain
Claims vs data	Each failure claim counted on 2,000 outputs before writing; two over-strong draft claims were corrected
Deliverables	Submission matches all 330 ids; Hugging Face weights match by SHA-256 and load with 3 dependencies

All 19 checks, with results, are in the appendix.

8 Next steps with more time or compute

- **Train to convergence** (dev still rising at 9k steps) with a cosine tail, and several seeds for variance.
- **Close the domain gap:** back-translate monolingual English literature (not opus_books) [12], domain tags, a copy mechanism for names.
- **Run the prepared ablations** (RoPE vs sinusoidal, 16k vs 32k vocab) and stronger data filtering [13].
- **Decoding:** KV cache plus MBR with chrF as the utility; report sacreBLEU alongside the challenge scorer.

References

- [1] Vaswani et al. 2017, Attention Is All You Need
- [2] Kasai et al. 2021, Deep Encoder, Shallow Decoder
- [3] Xiong et al. 2020, On Layer Normalization in the Transformer
- [4] Su et al. 2021, RoFormer: Rotary Position Embedding
- [5] Press et al. 2022, Train Short, Test Long (ALiBi)
- [6] Press & Wolf 2017, Using the Output Embedding
- [7] Kudo & Richardson 2018, SentencePiece
- [8] Kudo 2018, Subword Regularization
- [9] Sennrich et al. 2016, NMT of Rare Words with Subword Units
- [10] Wu et al. 2016, Google's NMT System
- [11] Popel & Bojar 2018, Training Tips for the Transformer
- [12] Sennrich et al. 2016, NMT with Monolingual Data
- [13] Junczys-Dowmunt 2018, Dual Conditional Cross-Entropy Filtering